

MS Thesis Research Proposal

Measuring Recency Bias in Evidence Utility Signals

Proposal updated: September 16, 2026

Abstract

Retrieval-augmented generation (RAG) systems admit passages to the context window based on relevance scores. These scores are typically derived from dense-passage retrievers trained on human relevance judgments. When the evidence undergoes clinical reassessment, older judgments are overrepresented in the training set relative to newer ones. This asymmetry in training data may cause the retriever to preferentially rank older evidence, introducing a systematic bias toward outdated clinical guidance.

This thesis investigates whether confidence-derived evidence admission filters exhibit recency asymmetry and whether explicitly incorporating evidence recency into the admission step reduces that asymmetry without degrading answer quality. Three primary arms are compared: an unfiltered control, a confidence baseline (Flan-T5 [HELPFUL]/[NOT_HELPFUL]), and a recency-aware admission policy. The proposed method combines rank-normalised reranker relevance with an explicit recency score: $A(s) = (1-\lambda) \cdot p(s) + \lambda \cdot R(s, q, t_q)$, where λ controls the recency weight and $R(s, q, t_q) = 2^{-age/H}$ is an exponential age-decay function.

The primary contribution is the Filter Recency-Bias (FRB) probe: a methodology for measuring asymmetric admission across temporal-counterfactual evidence pairs. The secondary contribution is a recency-aware admission policy that tests whether making evidence age visible to the admission step reduces the measured asymmetry without degrading answer quality.

The evaluation uses 512 temporally-paired clinical questions from an external dataset (MedChangeQA), with results reported on an Alzheimer's-focused retrieval corpus. Primary results are pre-registered; all analysis code, pair definitions, and evaluation sets are released.

Research Question

Does a confidence-derived evidence admission mechanism exhibit recency asymmetry, and does explicitly incorporating evidence recency reduce that asymmetry without simply degrading answer quality?

Primary Measurement

Admission asymmetry is measured as:

$$\Delta = P(\text{admit} \mid \text{older passage}) - P(\text{admit} \mid \text{newer passage})$$

where $\Delta > 0$ indicates older evidence is preferentially admitted. The matched temporal-counterfactual pair is the unit of analysis. This measurement is computed separately for each arm and compared across conditions.

Three Primary Arms

Arm	Admission Rule	Sees Dates?
B1_NO_FILTER	Admit everything, up to the budget	No
B2_RAG2	Flan-T5 [HELPFUL] / [NOT_HELPFUL]	No
P_RECENCY	$A(s) \geq \theta$, where A is recency-aware	Yes

All three arms receive the same question, candidate set (frozen, byte-identical), candidate order, context budget, prompt template, and decoding settings. Only the admission rule differs.

Recency-Aware Admission Policy

The proposed method combines two signals in a simple scoring rule:

$$A(s) = (1 - \lambda) \cdot \rho(s) + \lambda \cdot R(s, q, t_q)$$

Admit if $A(s) \geq \theta$

where:

- $\rho(s)$ is the rank-normalised reranker score (the signal the baseline already produces), in $[0, 1]$
- $R(s, q, t_q) = 2^{-\text{age_days} / H}$ is the recency score, where $\text{age_days} = t_q - \text{publication_date}(s)$, clamped at zero, in $(0, 1]$
- $\lambda \in [0, 1]$ controls how much of the score is recency (tuned on validation split)
- $\theta \in [0, 1]$ is the admission threshold (tuned on validation split)
- H is the half-life parameter in the age-decay function (tuned on validation split)

This is plain age decay and nothing else. The recency score has no learned weights, no non-linearities beyond the exponential, and no auxiliary metadata. The three tunable quantities are λ , θ , and H ; all are fitted on the validation split, never on test.

Ablations and Controls

Ablation ($\lambda = 0$): Pure relevance at the same θ , isolating what the recency signal contributes over thresholding relevance alone. This is reported as a sensitivity analysis, not as a fourth experimental arm.

Negative control (unchanged-claim condition): Same construction and matching, but the clinical claim did not change across the window. We expect $\Delta \approx 0$ there; $\Delta > 0$ on changed-claim pairs alongside $\Delta \approx 0$ on unchanged-claim pairs separates claim recency from prose-era style effects.

Expected Contributions

C1. Filter Recency-Bias (FRB) Probe Methodology

A measurement framework for detecting asymmetric admission of older vs. newer evidence in confidence-derived filters. Evaluation on temporally-paired clinical questions (MedChangeQA dataset) shows whether this class of filters systematically prefers outdated evidence. The unchanged-claim negative control rules out prose-era style effects. Results are generalizable across confidence-derived filters, not limited to this implementation.

C2. Recency-Aware Admission Policy

A two-signal scoring rule combining rank-normalised reranker relevance with an explicit recency score, evaluated against both the Flan-T5 confidence baseline and an unfiltered control. Tests whether making evidence age visible to the admission step reduces the measured asymmetry without degrading answer quality.

C3. Frozen Evaluation Artifacts

Released artefacts include the test-pair set with its exclusion log and attrition table, the frozen evaluation specification with its content hash (SHA-256), pre-registered analysis plan, and all analysis code. Enables independent replication and detection of any analysis changes after outcomes are observed.

Research Objectives

Objective 1: Establish baseline measurement of confidence-derived filter behaviour on temporally-paired evidence.

Objective 2: Quantify the magnitude of recency asymmetry in the baseline (Flan-T5) filter.

Objective 3: Implement and evaluate the recency-aware admission policy on the same evaluation set.

Objective 4: Compare admission asymmetry across the three arms and establish whether recency awareness reduces it without degrading answer quality.

Objective 5: Specify and implement a recency-aware admission policy combining reranker relevance with an explicit recency score.

Objective 6: Evaluate the recency-aware arm against the RAG² baseline and an unfiltered control.

Evaluation Plan

The evaluation uses 512 temporally-paired clinical questions from MedChangeQA (Vladika et al., Findings of EMNLP 2025), with evidence retrieved from an Alzheimer's-focused medical corpus. Each question has two passages: an older review reaching one verdict and a newer review reaching the opposite verdict. The information state (t_q) is set to the corpus snapshot date, which is later than every passage, so no evidence post-dates the evaluation moment.

Primary metrics are dual-reported: admission rates conditional on producing an answer, and admission rates with abstentions counted as failures. Three independent tuning/validation/test splits are used; all hyperparameters (λ , θ , H) are fitted on validation, evaluated on test.

Answer quality is measured using automatic metrics (BLEU, ROUGE, question-answering metrics). The thesis does not include a blinded clinician rating study; this moves to future work due to recruitment lead time. A small targeted clinician review of Alzheimer's case-study items may be included if feasible.

Primary Scope

This thesis addresses the core research question: does confidence-derived admission exhibit recency asymmetry, and does explicit recency awareness reduce it? The primary evaluation uses general-medical external-authored pairs; Alzheimer's evidence provides the retrieval corpus and case-study domain.

Explicitly out of scope: clinician rating study · comparison against further state-of-the-art filters · agentic reference arm · supersession table · claim-class-dependent inclusion rules · retrieval-quality optimisation · contested-evidence detection, supersession discounting, or entailment-derived support as primary mechanisms.

Secondary and Future Components

The repository includes implementations of secondary mechanisms (contested-evidence detection, supersession discounting, time-invariance, answer verification, entailment-derived support, source authority) off by default. Enabling any of them is a secondary analysis and must be declared as such; each run records which were active.

This proposal incorporates all scope amendments approved September 16, 2026. The frozen scope governs; see `FROZEN_SCOPE.md` and `PROPOSAL_SCOPE_AMENDMENT.md` in the repository.